LICENCE EN INFORMATIQUE • IFAG 2026

Mobile Match Algeria

A web platform for comparing Algerian mobile plans

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The problem

Three operators • ever-changing plans • no unified comparator

Mobilis
13 plans

Djezzy
33 plans

Ooredoo
37 plans

“Which plan really matches my budget and usage?”

Plans are scattered across three sites, change often, and there is no dedicated comparator for Algeria.

83 active plans • > 54 million mobile subscribers in Algeria (ARPCE, Q1 2025)

Objectives

Five objectives from the FICHE, all delivered

#	Objective	Delivered as
1	Responsive multi-operator comparator	`/offers` catalogue, filters, 83 live plans
2	Automated data aggregation	Playwright (headless Edge) + Cheerio, daily scrape
3	Intelligent recommendation engine	TOPSIS + rank-order-centroid weights
4	Comparison & savings visualisation	Side-by-side table, bar charts, savings indicator
5	In-app notification system	Two streams: user (offers) • admin (scrape)

System architecture

Three tiers in one Next.js codebase

Presentation

Next.js 16 (App Router) • React 19 • TypeScript • Recharts • Framer Motion

↓

Logic / API

Next.js route handlers • NextAuth (JWT, bcrypt) • recommendation engine • search parser • rate limiting

↓

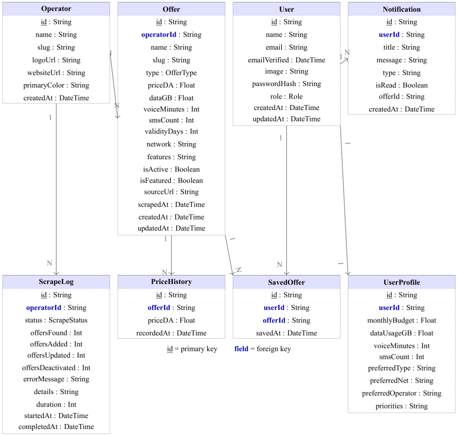
Data

Prisma 7 + SQLite • scraper orchestrator (Playwright + Cheerio)

Full-stack TypeScript — one language across UI and API. SQLite — zero-config, portable, sufficient at this scale.

Data model • class diagram

Eight domain entities • Prisma schema = single source of truth



id = primary key • **field** = foreign key • NextAuth / infra tables omitted

Concept 1 — Web data aggregation

Tool: **Playwright (headless Edge) + Cheerio**

Why it's the best choice

Option	Verdict
Public API / feed	None published by the operators
Scrapy (no browser)	Blocked by anti-bot
OCR (Tesseract)	29–43% confidence — rejected
Playwright + Cheerio	Real browser defeats anti-bot; clean HTML

Hierarchy: API > feed > scraping (Boegershausen & Datta, 2022).

How it works

Operator websites
↓
Playwright • headless Edge
↓
Cheerio parser
↓
validate.ts • zod (12 rules)
↓
Prisma → SQLite

Example: daily cron at 03:00 → 83 plans normalised to one schema (MB→GB, prices, defaults) — catalogue 100% live.

Concept 2 — Recommendation engine

Tool: TOPSIS + rank-order-centroid weights

Why it's the best choice

Method	Verdict
Collaborative / ML	No interaction history (cold start)
Weighted sum	Unbounded, scale-sensitive
Lexicographic	Too rigid (ignores trade-offs)
AHP	Too many pairwise comparisons
TOPSIS + ROC	Content-based, bounded score, every step published

How it works

Hard filters (conjunctive)
budget • operator • type • network

↓

ROC weights from the user's ranking
price#1, data#2 $\rightarrow$ 0.75 / 0.25

↓

TOPSIS closeness to the ideal

$$C = \frac{S^-}{S^+ + S^-} \in [0, 1]$$

↓

Top 3 plans by C

Hwang & Yoon (1981) • Barron & Barrett (1996) — nothing in the ranking is invented.

Recommendation — worked example

Price scored as *value for money* (DA per GB), not raw price

Plan	Price	Data	DA / GB	
A	600 DA	30 GB	**20**	**best value**
B	2000 DA	50 GB	40	most data
C	400 DA	8 GB	80	cheapest, worst value

Priority = price → A
best value, *not* the cheapest C

Priority = data → B
the most data

The cheapest plan (C) ranks *last* — “price first” rewards value for money, not a low sticker price.

Concept 3 — Search engine

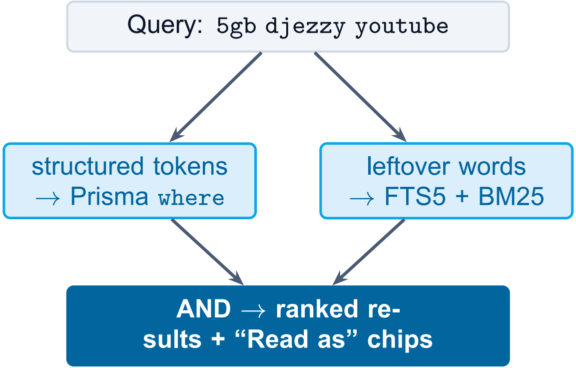
Tool: **token parser + SQLite FTS5 (BM25)**

Why it's the best choice

Option	Verdict
One big `LIKE`	Cannot rank, slow
External engine	Extra service, 2nd source of truth
Parser + FTS5	Exact filters + BM25 relevance, in the DB file

FTS5 ships inside SQLite: BM25 ranking, prefix match, accent folding.

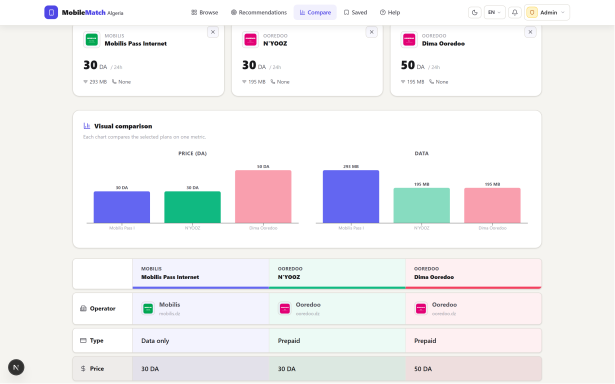
How it works



Example: `youtube` matches no token → FTS5 finds plans whose features mention it; the matched feature is highlighted on the card.

Concept 4 — Comparison & visualisation

Tool: comparison table + bar charts (Recharts) + savings



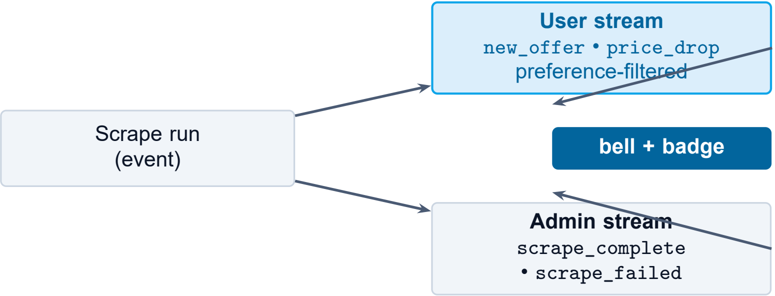
Why it's the best choice

- Side-by-side **table** — the standard pattern for structured products.
- **Bar charts** use position + length, the most accurate visual encoding for quantities (Munzner, 2014).
- **Savings** shown per plan: monthly, annual, % vs the user's budget.

Example: up to 3 plans, tinted by operator; price / data / annual-savings bar charts.

Concept 5 — Notification system

Tool: publish / subscribe, two separate streams



Why two streams?

Each audience sees only what concerns it — a shopper never sees scrape logs, an admin never sees price-drop alerts (publish/subscribe — Eugster et al., 2003).

Why preference-filtered?

Matching each alert to a declared interest keeps the attention cost low and avoids over-notification (McCrickard & Chewar, 2003).

Engineering & validation

Tested, type-safe, defensible

Technology stack

Next.js 16	full-stack React framework
Prisma 7	type-safe ORM, schema = truth
SQLite	zero-config, portable
NextAuth	JWT sessions, bcrypt, roles
Playwright	headless Edge, anti-bot
Vitest	unit + snapshot tests

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unit tests passing

Engine • scraping parsers • search parser • helpers.

+ multi-account isolation test: no user can read another's profile, saved list, or notifications.

Limitations $\rightarrow$ perspectives

What is honestly missing today, and what closes the gap

Limitation today	Perspective
No formal user-survey study	Usability study with external testers (SUS)
Scraper fragility on redesigns	Snapshot tests + circuit breaker $\rightarrow$ richer monitoring
Image-only plan families excluded	Cloud Vision / LLM extraction
Unlimited calls/SMS modelled as binary	Refine to a scoped attribute (on-net vs all)
Single-machine evaluation	Public cloud deployment (no code change)

Conclusion

The first dedicated **mobile-plan comparator for Algeria** — automated aggregation, a researched recommendation engine, structured search, and clear visual comparison.

83
live plans, 3 operators

5 / 5
objectives delivered

121
tests passing

Every tool choice is backed by the literature; the recommendation ranking uses only published formulas.

MOBILE MATCH ALGERIA

Thank you.

Questions and discussion welcome.

Live demo   http://localhost:3000

Source   github.com/Riad-Amt00/Mobile-Match-Algeria

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